Experiment 8: Time Series Prediction using RNN.

```
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split

def generate_time_series(n_steps=1000):
  time = np.linspace(0, 50, n_steps)
  series = np.sin(time) + 0.3 * np.sin(3 * time) + 0.2 *
np.random.randn(n_steps)
  return series.reshape(-1, 1)

data = generate_time_series(1000)

scaler = MinMaxScaler(feature_range=(-1, 1))
data_normalized = scaler.fit_transform(data)

def create_sequences(data, seq_length=20):
  X, y = [], []
  for i in range(len(data) - seq_length):
    X.append(data[i:i+seq_length])
    y.append(data[i+seq_length])
  return np.array(X), np.array(y)

SEQ_LENGTH = 20
X, y = create_sequences(data_normalized, SEQ_LENGTH)

X_train, X_test, y_train, y_test = train_test_split(
  X, y, test_size=0.2, random_state=42, shuffle=False
)

X_train = torch.FloatTensor(X_train)
y_train = torch.FloatTensor(y_train)
X_test = torch.FloatTensor(X_test)
y_test = torch.FloatTensor(y_test)

print(f"X_train shape: {X_train.shape}")
print(f"y_train shape: {y_train.shape}")

class TimeSeriesLSTM(nn.Module):
  def __init__(self, input_size=1, hidden_size=64, num_layers=2,
output_size=1):
    super(TimeSeriesLSTM, self).__init__()
    self.hidden_size = hidden_size
    self.num_layers = num_layers

    self.lstm = nn.LSTM(
      input_size=input_size,
      hidden_size=hidden_size,
      num_layers=num_layers,
      batch_first=True,
      dropout=0.2
    )
    self.fc = nn.Linear(hidden_size, output_size)

  def forward(self, x):
    h0 = torch.zeros(self.num_layers, x.size(0), self.hidden_size)
    c0 = torch.zeros(self.num_layers, x.size(0), self.hidden_size)
    out, _ = self.lstm(x, (h0, c0))
    out = self.fc(out[:, -1, :])
    return out

model = TimeSeriesLSTM(input_size=1, hidden_size=64, num_layers=2,
output_size=1)
criterion = nn.MSELoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

print(f"Model parameters: {sum(p.numel() for p in
model.parameters())}")

def train_model(model, X_train, y_train, X_test, y_test, epochs=100):
  train_losses = []
  test_losses = []

  for epoch in range(epochs):
    model.train()
    optimizer.zero_grad()

    train_outputs = model(X_train)
    train_loss = criterion(train_outputs, y_train)

    train_loss.backward()
    optimizer.step()

    model.eval()
    with torch.no_grad():
      test_outputs = model(X_test)
      test_loss = criterion(test_outputs, y_test)

    train_losses.append(train_loss.item())
    test_losses.append(test_loss.item())

    if (epoch + 1) % 20 == 0:
      print(f'Epoch [{epoch+1}/{epochs}], Train Loss:
{train_loss.item():.4f}, Test Loss: {test_loss.item():.4f}')

  return train_losses, test_losses

print("\nTraining...")
train_losses, test_losses = train_model(model, X_train, y_train, X_test,
y_test, epochs=150)

model.eval()
with torch.no_grad():
  train_predictions = model(X_train).numpy()
  test_predictions = model(X_test).numpy()

train_predictions = scaler.inverse_transform(train_predictions)
test_predictions = scaler.inverse_transform(test_predictions)
y_train_actual = scaler.inverse_transform(y_train.numpy())
y_test_actual = scaler.inverse_transform(y_test.numpy())

plt.figure(figsize=(15, 10))

plt.subplot(2, 2, 1)
plt.plot(train_losses, label='Train Loss')
plt.plot(test_losses, label='Test Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Training and Testing Loss')
plt.legend()
plt.grid(True)
```

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plt.subplot(2, 2, 2)
plt.plot(y_test_actual, label='Actual', alpha=0.7)
plt.plot(test_predictions, label='Predicted', alpha=0.7)
plt.xlabel('Time Step')
plt.ylabel('Value')
plt.title('Test Set: Actual vs Predicted')
plt.legend()
plt.grid(True)

plt.subplot(2, 2, 3)
plt.scatter(y_test_actual, test_predictions, alpha=0.5)
plt.plot([y_test_actual.min(), y_test_actual.max()],
      [y_test_actual.min(), y_test_actual.max()], 'r--', lw=2)
plt.xlabel('Actual Values')
plt.ylabel('Predicted Values')
plt.title('Prediction vs Actual (Test Set)')
plt.grid(True)

plt.subplot(2, 2, 4)
full_predictions = np.concatenate([train_predictions, test_predictions])
plt.plot(scaler.inverse_transform(data_normalized), label='Original
Data', alpha=0.7)
plt.plot(range(SEQ_LENGTH, SEQ_LENGTH + len(full_predictions)),
      full_predictions, label='Predictions', alpha=0.7)
plt.xlabel('Time Step')
plt.ylabel('Value')
plt.title('Full Time Series with Predictions')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()

def predict_future(model, last_sequence, n_steps=10):
  model.eval()
  current_sequence = last_sequence.clone().detach().float()
  predictions = []

  with torch.no_grad():
    for _ in range(n_steps):
      next_pred = model(current_sequence.unsqueeze(0))
      predictions.append(next_pred.item())
      current_sequence = torch.cat([
        current_sequence[1:],
        torch.FloatTensor([[next_pred.item()]])
      ])

  return np.array(predictions).reshape(-1, 1)

last_sequence = X_test[-1]

future_steps = 30
future_predictions = predict_future(model, last_sequence,
future_steps)
future_predictions = scaler.inverse_transform(future_predictions)

plt.figure(figsize=(12, 5))
plt.plot(range(len(y_test_actual)), y_test_actual, label='Test Data',
alpha=0.7)
plt.plot(range(len(y_test_actual), len(y_test_actual) + future_steps),
      future_predictions, 'r--', label='Future Predictions', alpha=0.7)
plt.xlabel('Time Step')
plt.ylabel('Value')
plt.title(f'Future {future_steps}-Step Predictions')
plt.legend()
plt.grid(True)
plt.show()

from sklearn.metrics import mean_absolute_error,
mean_squared_error, r2_score

mae = mean_absolute_error(y_test_actual, test_predictions)
mse = mean_squared_error(y_test_actual, test_predictions)
rmse = np.sqrt(mse)
r2 = r2_score(y_test_actual, test_predictions)

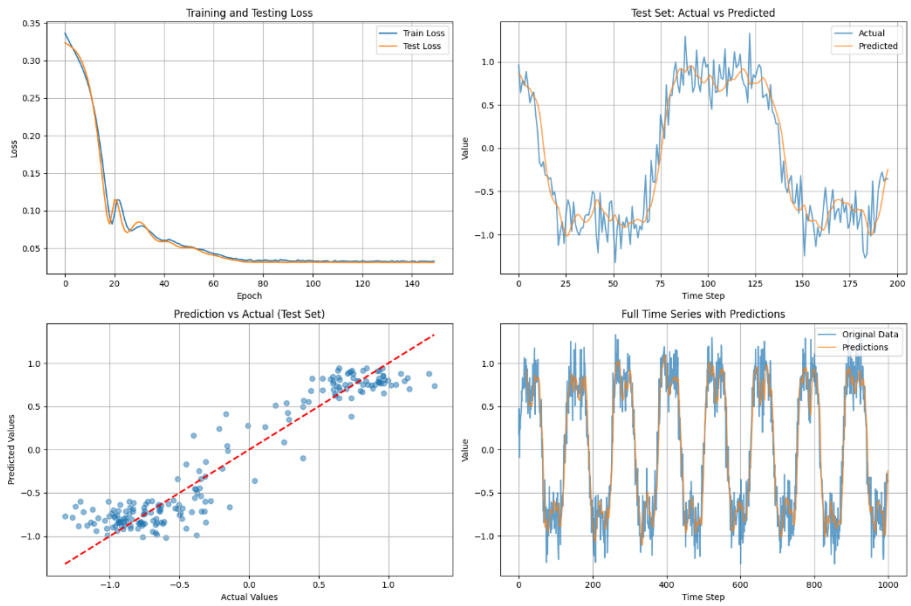
print("\n" + "="*50)
print("MODEL PERFORMANCE METRICS")
print("="*50)
print(f"Mean Absolute Error (MAE): {mae:.4f}")
print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.4f}")
print(f"R² Score: {r2:.4f}")
print("="*50)
```

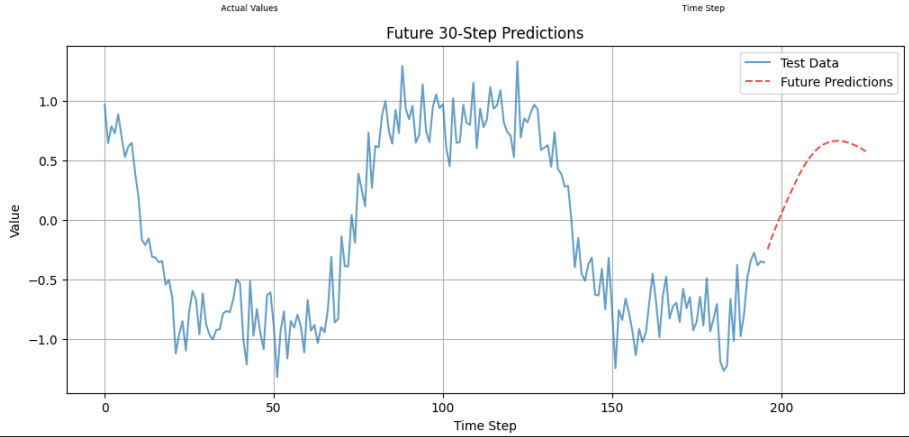
Observation

The RNN (LSTM) model successfully learned the temporal patterns of the time series data and predicted future values based on past sequences. Training and testing loss decreased over epochs, indicating effective learning. The predicted values closely followed the actual data trend, though minor deviations occurred due to noise. Overall, the model demonstrated good capability for time series forecasting.

```
X_train shape: torch.Size([784, 20, 1])
y_train shape: torch.Size([784, 1])
Model parameters: 50497

Training...
Epoch [20/150], Train Loss: 0.0820, Test Loss: 0.0977
Epoch [40/150], Train Loss: 0.0604, Test Loss: 0.0586
Epoch [60/150], Train Loss: 0.0438, Test Loss: 0.0412
Epoch [80/150], Train Loss: 0.0335, Test Loss: 0.0314
Epoch [100/150], Train Loss: 0.0334, Test Loss: 0.0314
Epoch [120/150], Train Loss: 0.0318, Test Loss: 0.0310
Epoch [140/150], Train Loss: 0.0317, Test Loss: 0.0311
```






```
==================================================
MODEL PERFORMANCE METRICS
==================================================
Mean Absolute Error (MAE): 0.1848
Mean Squared Error (MSE): 0.0544
Root Mean Squared Error (RMSE): 0.2332
R² Score: 0.9048
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```